
PDS Slow Controls Vertical Slice

What has been proven at CERN VST,
and what remains in the development plan

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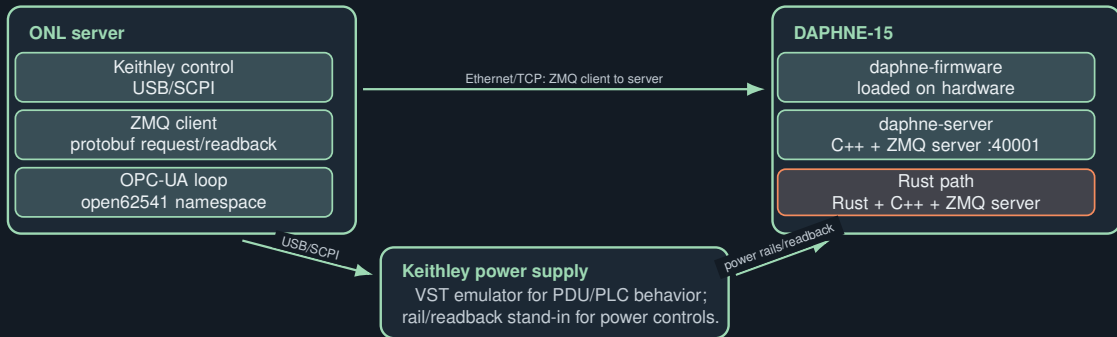


CERN VST Proof of Concept Boundary

HARDWARE FIRST: THREE EQUIPMENT ITEMS IN THE TEST

green = tested at CERN VST

orange = planned or not tested in this slice



Tested at CERN VST: ONL runs the Keithley control, ZMQ client, and OPC-UA loop; DAPHNE runs firmware plus the legacy C++ ZMQ server. The final PDU/PLC/interlock replacement and Rust/RPU server path remain planned.

Development Gaps Before Production

WHAT THE VST PROOF DOES NOT YET DEFINE

Interfaces to freeze

Hardware database

SC schema and versioned JSON exports for device, endpoint, rack/slot, IP, outlet, owner, and dependency edges.

OPC-UA namespaces

Generated Nodelds, units, limits, quality, timestamps, ownership, and alarm metadata; no hand-maintained map.

Missing contracts

ZMQ/protobuf schemas, command acknowledgements, error taxonomy, retry rules, lifecycle states, and health readbacks.

Control path still open

Interlocks and sensors

PDU/PLC state, smoke sensors, trip latches, bounds, write-enable logic, and fail-closed behavior.

Migration layers

C++ reference path, Rust candidate path, and RPU-owned AFE access need shadow-mode tests before writes.

Integration holes

Ignition import, historian policy, alarm classes, deployment, credentials, subnets, and power-cycle recovery.

Development Plan: Keep the Interface Stable

THE PLAN IS AN INTEGRATION SEQUENCE, NOT A REWRITE



Sources: github.com/marroyav/daphne-sc for the tested C++/OPC-UA vertical slice;
github.com/marroyav/daphne_slow_controls_plan for topology and SC workflow planning.

Rust vs C++: Decisions Still Open

C++ IS THE LIVE HARDWARE REFERENCE; RUST IS THE COMPATIBILITY AND RPU-SAFETY PATH

Tested reference

- ▶ Legacy C++ `daphneServer` replies on DAPHNE-15 40001.
- ▶ `ControlEnvelopeV2` readbacks are the current compatibility contract.
- ▶ C++ OPC-UA bridge reads live values on ONL.

Rust target

- ▶ Protocol-compatible server on a separate endpoint until parity.
- ▶ Linux aggregates status and enforces preflight.
- ▶ AFE-affecting commands fail closed without a valid RPU backend.

Open questions

- ▶ What response parity lets Rust replace or front C++?
- ▶ Does OPC-UA remain a small C++ bridge or move into Rust?
- ▶ What RPU ABI test is required before any AFE write migration?

Boundary: C++ legacy service is the tested hardware reference; Rust is for protocol-compatible service evolution and RPU-owned safety.

Slow Controls Workflow Started Today

WORKFLOW FIRST: INVENTORY, EXPORT, GENERATE, THEN CONNECT



What we have

- ▶ Draft SC plan and topology view.
- ▶ DAPHNE-15 VST path through C++ server and OPC-UA bridge.
- ▶ OPC-UA smoke read: heartbeat, VBIAS, rails, and summary.

What is missing

- ▶ Final HWDB SC schema and exports for all subsystems.
- ▶ PDU, smoke, interlock, Gizmo, CRP top/bottom, and PDS interfaces.
- ▶ Ignition tag model, alarms, historian, and operator screens.

Immediate Work Packages

NEXT: KEEP THE GREEN PATH STABLE; TURN ORANGE ITEMS GREEN

1. Harden the VST slice

Reviewed ONL OPC-UA bridge;
DAPHNE remains read-only until
ownership is settled.

2. Extend the device model

PDU outlet map, smoke status, and
interlock nodes before write paths.

3. Close implementation choices

Rust/C++ parity on DAPHNE-15;
RPU ABI test before AFE writes.

CERN VST has proven the read-only DAPHNE-to-OPC-UA path. The full SC topology remains orange until HWDB exports, PDU/interlocks, Rust/RPU ownership, and Ignition are exercised.